CYTOSKELETON BDS

HMH

Learning Objectives

By the end of the lecture, students will be able to

- define and classify cytoskeleton with examples
- discuss general functions of cytoskeleton in terms of structure, support and functions
- discuss each class, microfilaments, intermediate filaments and microtubules w.r.t shape, site and functions
- relate diseased condition

- **Definition**: The cytoskeleton is a network of fibers throughout the cell's cytoplasm that helps the cell maintain its shape and gives support to the cell.

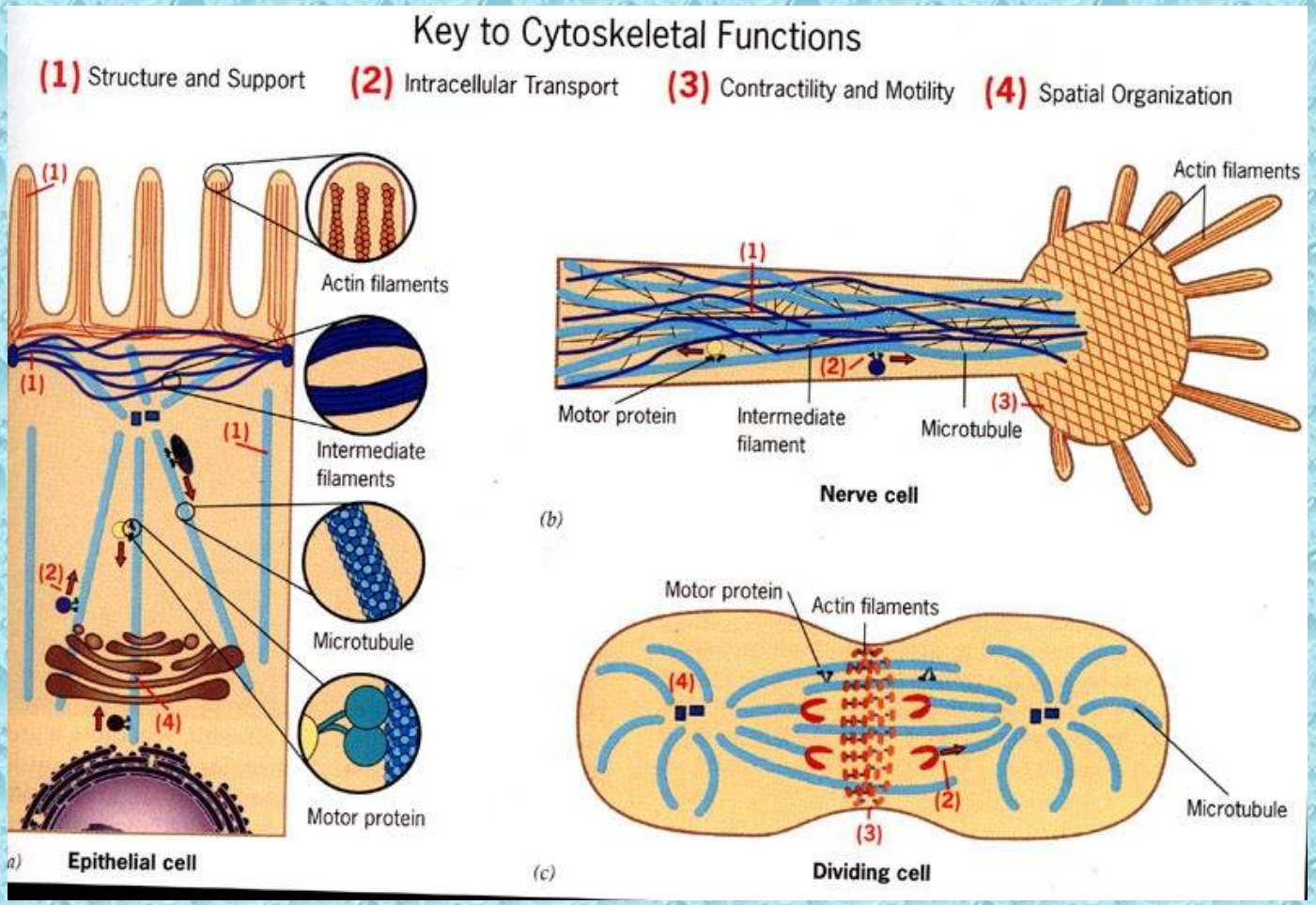
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Cell shape. For cells without cell walls, the cytoskeleton determines the shape of the cell. This is one of the functions of the intermediate filaments.

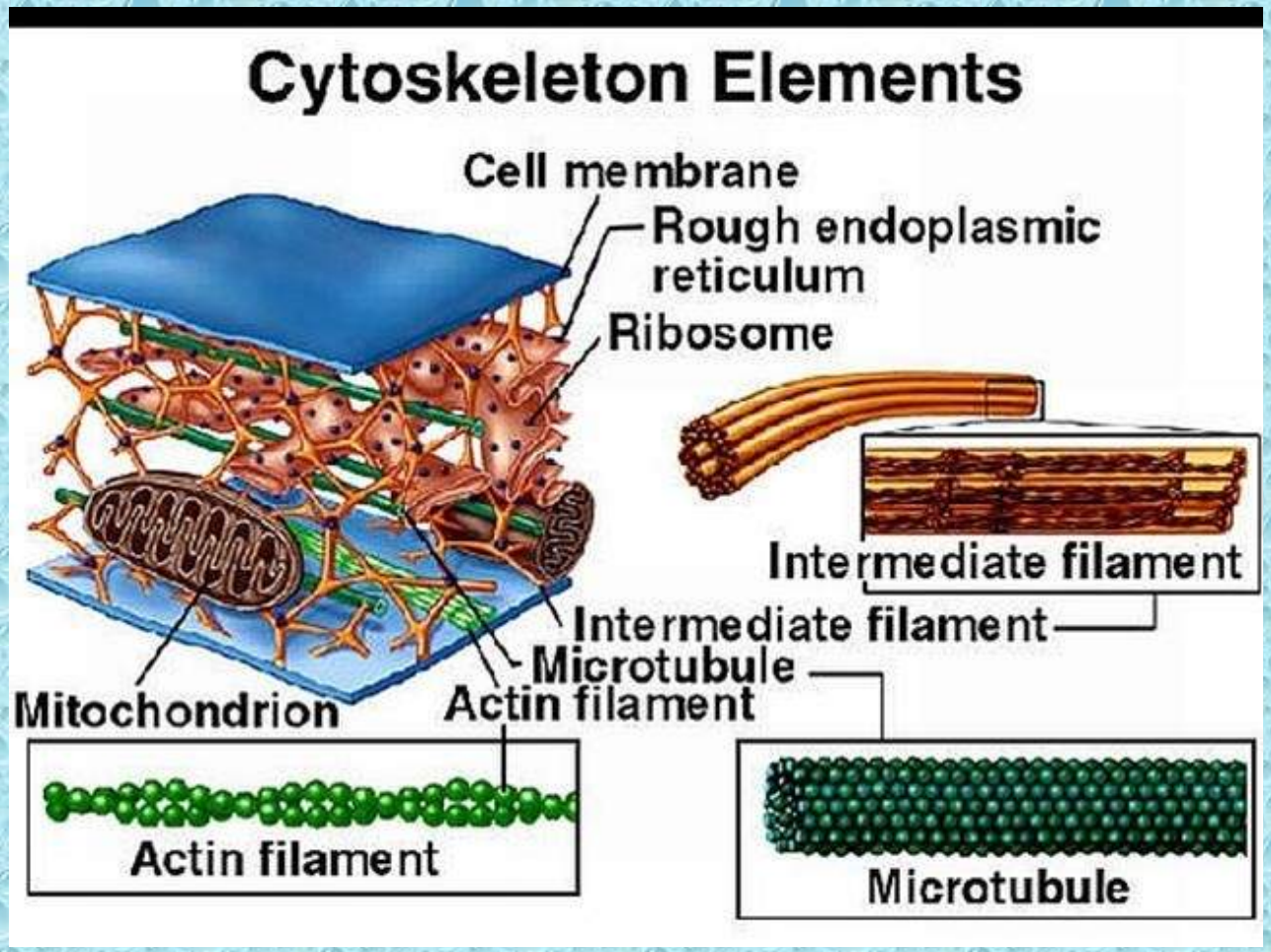
Cell movement. The dynamic collection of microfilaments and microtubles can be continually in the process of assembly and disassembly, resulting in forces that move the cell. There can also be sliding motions of these structures. Audesirk and Audesirk give examples of white blood cells "crawling" and the migration and shape changes of cells during the development of multicelled organisms

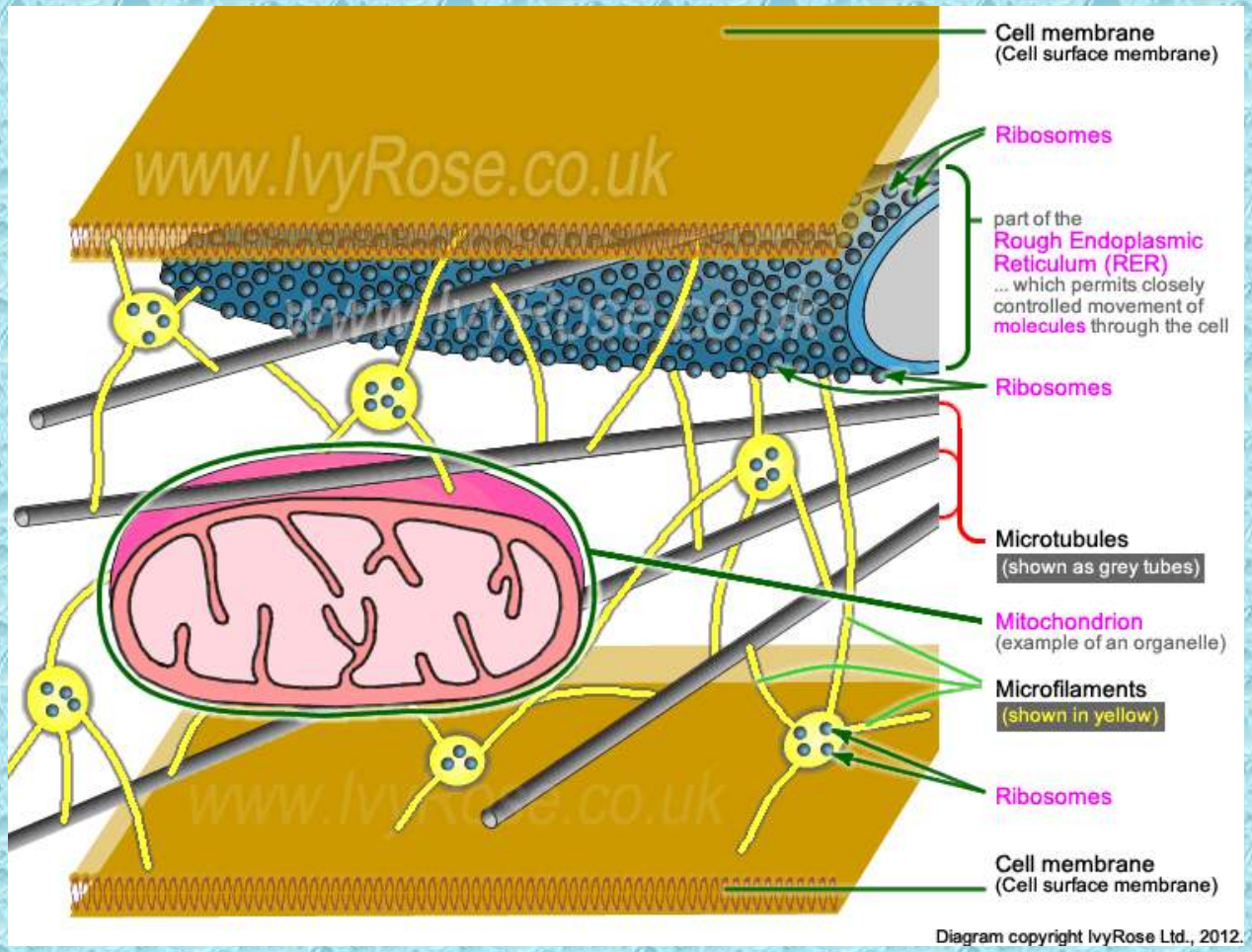
Cell division. During cell division, microtubules accomplish the movement of the chromosones to the daughter nucleus. Then for animal cells, a ring of microfilaments helps divide the two developing cells by constricting the central region between the cells.

Organelle movement. Microtubules and microfilaments can help move organelles from place to place in the cell. In the case of endocytosis when a vesicle is formed to engulf some particle from outside the cell, microfilaments attach to the vesicle and pull it into the cell. Much of the complex synthesis and distribution function of the endoplasmic reticulum and the Golgi complex makes use of transport vescicles, and it is thought that the movement of these vesicles is guided by the cytoskeleton.


Key to Cytoskeletal Functions
(1) Structure and Support
(2) Intracellular Transport
(3) Contractility and Motility
(4) Spatial Organization
(1)
Actin filaments
(1)
(1)
Intermediate filaments
(2)
Microtubule
(4)
Motor protein
(a) Epithelial cell
Actin filaments
(1)
(2)
(3)
Motor protein
Intermediate filament
Microtubule
Nerve cell
(b)
Motor protein
Actin filaments
(4)
(2)
(3)
Microtubule
(c) Dividing cell

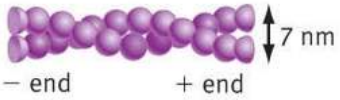
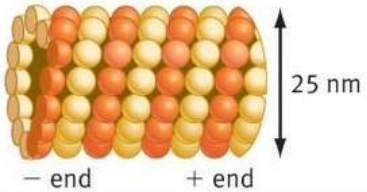
Cytoskeleton structure




Cell membrane
(Cell surface membrane)
Ribosomes
part of the
Rough Endoplasmic
Reticulum (RER)
... which permits closely
controlled movement of
molecules through the cell
Ribosomes
Microtubules
(shown as grey tubes)
Mitochondrion
(example of an organelle)
Microfilaments
(shown in yellow)
Ribosomes
Cell membrane
(Cell surface membrane)
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SUMMARY TABLE 7.3 **Cytoskeletal Filaments**

The three types of filaments found in the cytoskeleton are distinguished by their size and structure, and the protein subunit of which they are made.

	Structure	Subunits	Functions
Actin filaments (microfilaments)	Strands in double helix 	Actin 	• maintain cell shape by resisting tension (pull) • move cells via muscle contraction or cell crawling • divide animal cells in two • move organelles and cytoplasm in plants, fungi, and animals
Intermediate filaments	Fibers wound into thicker cables 	Keratin or vimentin or lamin or others 	• maintain cell shape by resisting tension (pull) • anchor nucleus and some other organelles
Microtubules	Hollow tube 	α- and β-tubulin dimers 	• maintain cell shape by resisting compression (push) • move cells via flagella or cilia • move chromosomes during cell division • assist formation of cell plate during plant cell division • move organelles • provide tracks for intracellular transport

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Property	Microtubules	Microfilaments (Actin Filaments)	Intermediate Filaments
Structure	Hollow tubes; wall consists of 13 columns of tubulin molecules	Two intertwined strands of actin	Fibrous proteins supercoiled into thicker cables
Diameter	25 nm with 15-nm lumen	7 nm	8–12 nm
Protein subunits	Tubulin, consisting of α-tubulin and β-tubulin	Actin	One of several different proteins of the keratin family, depending on cell type
Main functions	Maintenance of cell shape (compression-resisting "girders") Cell motility (as in cilia or flagella) Chromosome movements in cell division Organelle movements	Maintenance of cell shape (tension-bearing elements) Changes in cell shape Muscle contraction Cytoplasmic streaming Cell motility (as in pseudopodia) Cell division (cleavage furrow formation)	Maintenance of cell shape (tension-bearing elements) Anchorage of nucleus and certain other organelles Formation of nuclear lamina

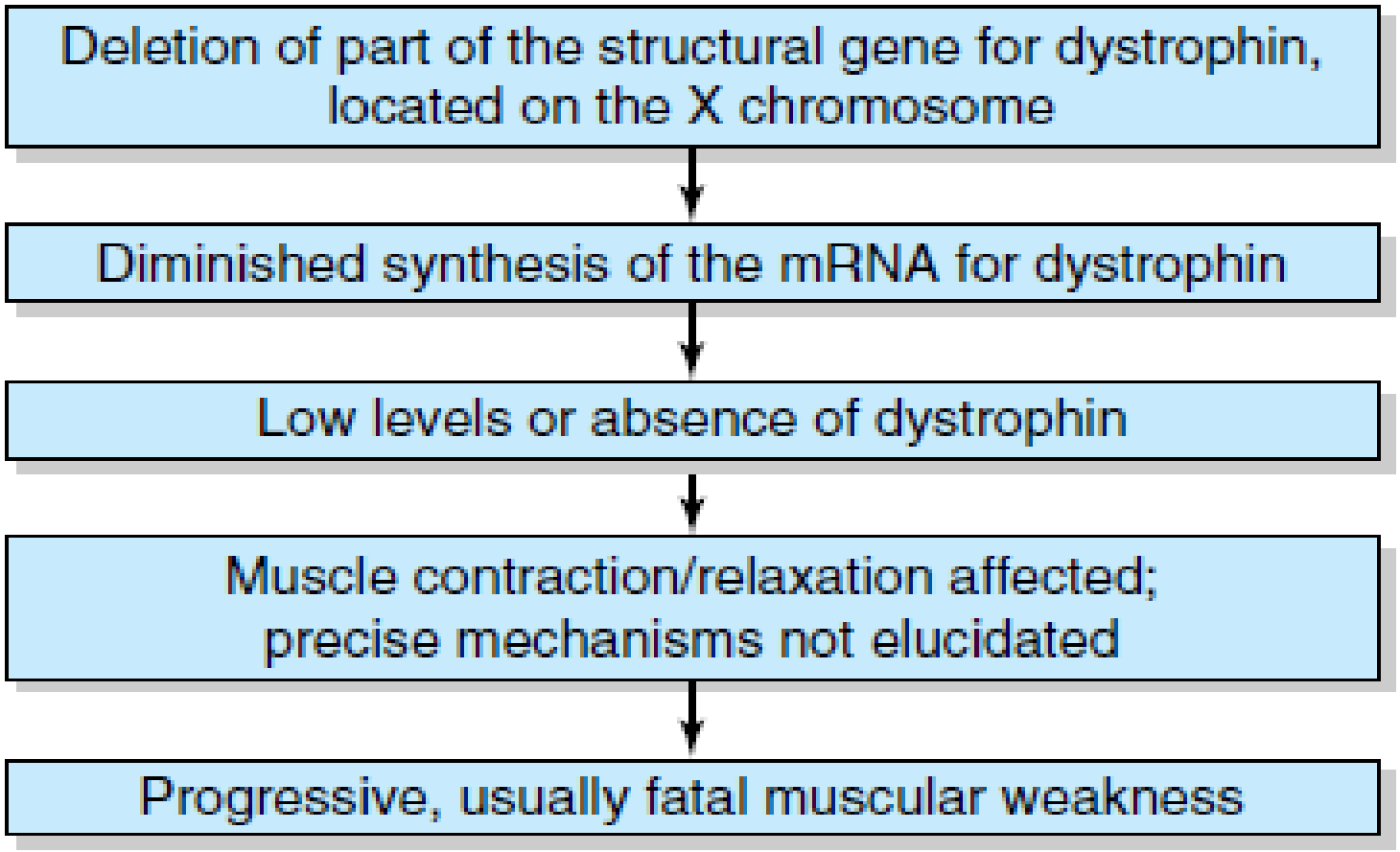


Figure 49–11. Summary of the causation of Duchenne muscular dystrophy (MIM 310200).

- Dystrophin links the actin cytoskeleton to the ECM and appears to be needed for assembly of the synaptic junction.

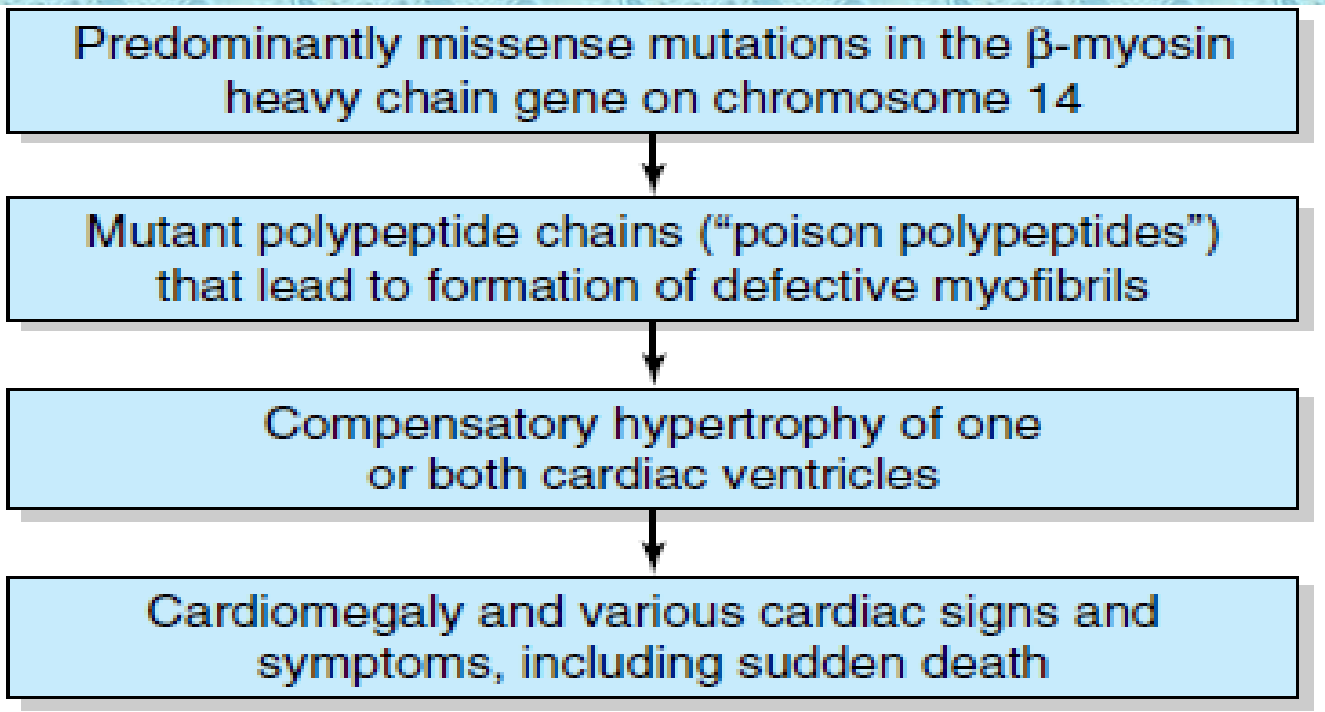


Figure 49–13. Simplified scheme of the causation of familial hypertrophic cardiomyopathy (MIM 192600) due to mutations in the gene encoding β-myosin heavy chain. Mutations in genes encoding other proteins, such as the troponins, tropomyosin, and cardiac myosin-binding protein C can also cause this condition. Mutations in genes encoding yet other proteins (eg, dystrophin) are involved in the causation of dilated cardiomyopathy.

An absence of dynein in cilia and flagella results in immotile cilia and flagella, leading to male sterility and chronic respiratory infection, a condition known as **Kartagener syndrome.**

Certain **drugs** bind to microtubules and thus interfere with their assembly or disassembly. These include colchicine (used for treatment of acute gouty arthritis), vinblastine (a vinca alkaloid used for treating certain types of cancer), paclitaxel (Taxol) (effective against ovarian cancer), and griseofulvin (an antifungal agent).

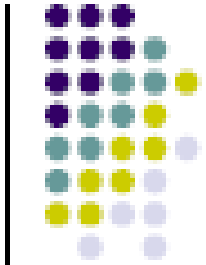
Cell junctions

- Plasma membrane areas specialized to provide contact between cells
  - Dense clusters of cell adhesion molecules on the outside
  - linked to cytoskeleton on the inside
  - through adapter proteins
- Four classes of cell junctions

Cell junctions – types based on localization

- Between cells
  - (1) tight junctions
  - (2) adherens junctions and desmosomes
  - (3) gap junctions
- Between cells and matrix
  - (4) hemidesmosomes

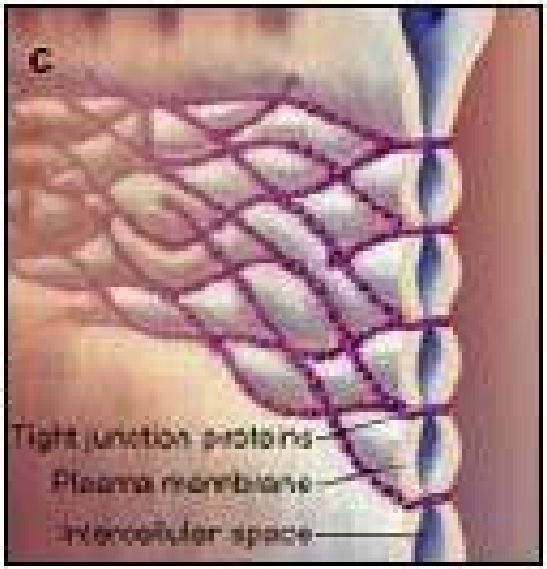
Cell junctions - types based on function



- Adhering junctions
  - (1) tight junctions
  - (2) adherens junctions and desmosomes
  - (4) hemidesmosomes
- Communicating junctions
  - (3) gap junctions

Tight junctions (zonula occludens)

- Belts of proteins that close extracellular space between cells
- Prevent passage of water and water-soluble substances
- Account for electrical resistance across epithelia



Regulation of tight junctions

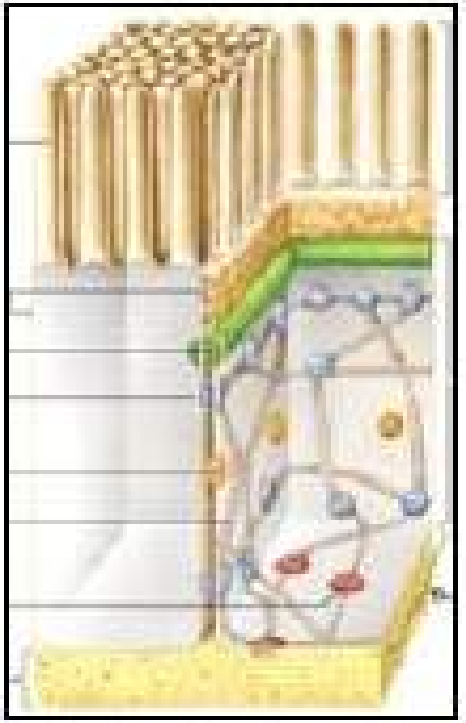
- Hormones
  - Vasopressin
- Cytokines
- Lack of ATP causes "leak"
- Extravasating leukocytes open tight junctions

Adhesive junctions

- Adherens junctions and desmosomes
- Hold cells tightly together
- Confer mechanical strength
- Common in tissue that are subject to severe stress such as skin and cardiac muscle

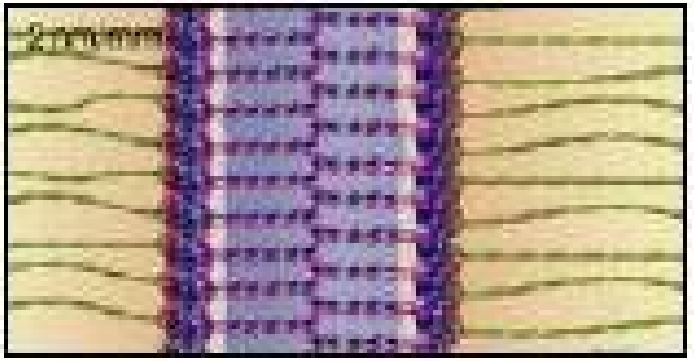
Molecular structure of adherens junctions

- Belt like junctions located just below tight junction
- Simple points of attachment, do not contain channels connecting the interiors of the two attached cells



Desmosomes

- Button like welds joining opposing cell membranes
- Cadherins bind the membranes of adjacent cells in a way that gives strength and rigidity to the entire tissue



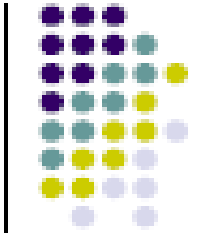
Gap junctions

- Junctions that provide direct connections (door) between cells
- Channels or pores through the membranes of two cells and across the intercellular space
- Form electrical synapses
  - Direct transmission of action potential without transmitter, receptors etc

Gap junctions

- Integrate the metabolism of the cells
- Metabolic coupling or metabolic cooperation – transfer of molecules that the neighboring cell can not synthesize (second messengers, precursors of DNA and RNA)

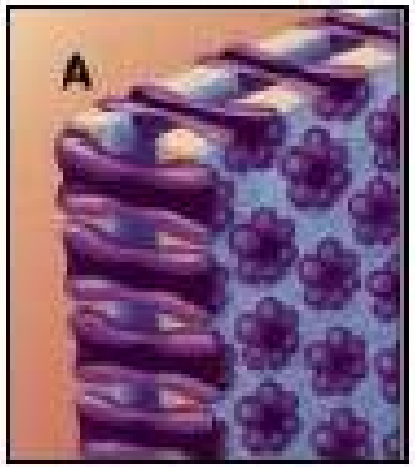
Cells that use gap junctions



- Skin epithelium
- Endocrine glands
- GI epithelium
- Smooth muscle
- Cardiac muscle
- Osteocytes
- Glial cells

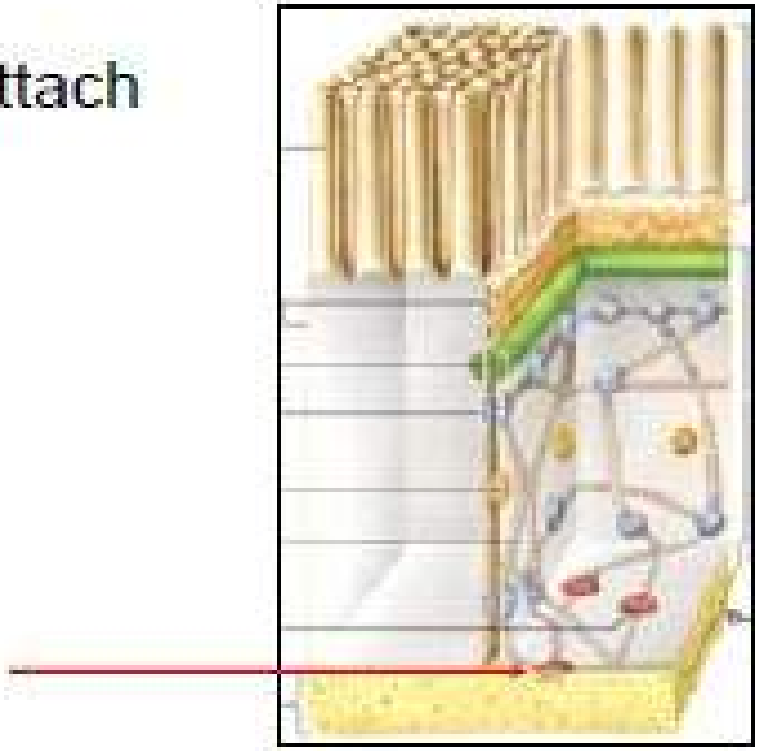
Molecular structure of gap junction

- A ring of 6 membrane proteins called connexins - connexons
- Two connexons on neighboring membranes form a transmembrane channel that interconnects the cytoplasms of two cells
- Connexons are size filters



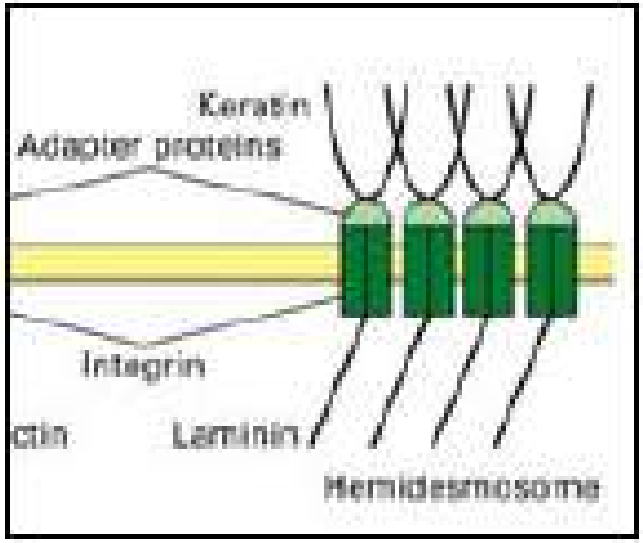
Hemidesmosomes

- Similar to desmosomes but totally different molecular structure
- Cell-matrix adhesions – attach cells to basal lamina



Molecular structure of hemidesmosomes

- Composed of integrins (outside) that bind to type XVII collagen and laminin-5
- Cytosolic side consist of a plague composed of adapter proteins (plectin) attaching integrins to keratin filaments



References

- The Cell
- Harpers Biochemistry